

Chapter 2

Transcription factor-DNA interactions

Regulatory proteins and other small molecules are constantly approaching genomic DNA. The significant players in transcriptional regulation are transcription factors (TFs). Without TFs, RNA polymerase and the basic transcriptional machinery would be unable to approach gene transcription start sites and begin transcription. TFs act in different modes and can find and bind regulatory regions: promoters and enhancers. Bound TFs attract Mediator proteins and Pol II and form a loop that brings enhancers and promoter regions close in space even though they can be separated by kilobases in a linear space (Figure 2.5). The activation of gene expression is tightly regulated and is cell type- and time-specific. Cooperation between two or more TFs is essential to ensure the complexity of transcription regulation. This chapter will overview the cis-regulatory regions, transcription factors, and interactions between the two.

2.1 Cis-regulatory regions

Regulatory regions in the eukaryotic genome are DNA segments associated with the capability to increase or decrease transcription. These regions are termed cis-regulatory regions (CRRs). They are crucial for gene expression regulation because they contain short TF-targeted DNA sequences called transcription factor binding sites (TFBSs) [22]. CRRs include promoters, enhancers, and silencers.

2.1.1 Promoters

Promoters are located at the 5' end region of the gene (Figure 2.5A) and are essential for gene expression regulation. Core promoters are located 40 bp upstream or downstream from the transcription start site (TSS) and are minimal regions needed for transcription initiation [23, 24]. Promoters contain specific DNA sequence elements essential for gene regulation. For example, TBP binds the TATA box element in the core promoter. The Initiator sequence is one of the most common elements and encompasses the TSS [23, 25]. Sequence-specific TFs can bind their sequence motifs in promoters [22–25]. Another characteristic common for human promoters is the abundance of CpG dinucleotides. CpGs are targets for cytosine methylation, one of the gene regulatory mechanisms [22, 26].

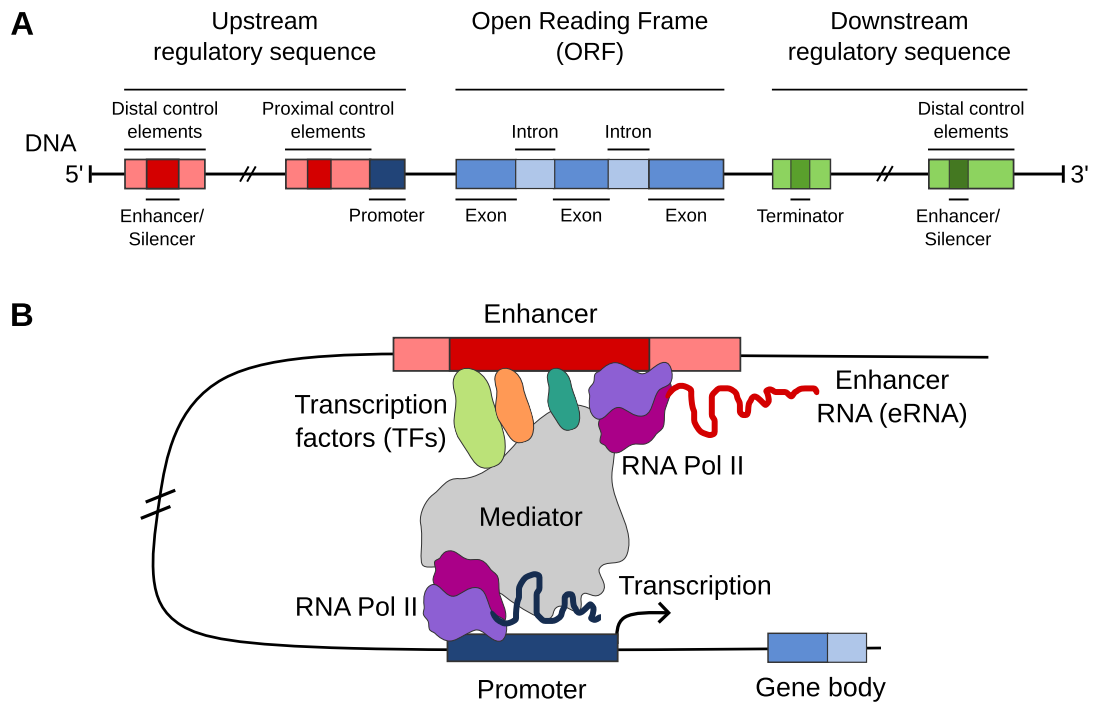


Figure 2.5: Transcription is regulated through protein binding to the gene cis-regulatory regions. **A.** Upstream and downstream of the gene body (open reading frame) lie regulatory regions. Regulatory sequences can be nearby (proximal) or separated by numerous bases (distal). Distal regulatory regions are enhancers and silencers. **B.** For transcriptional machinery to know which gene to transcribe, transcription factors (TFs) are binding enhancers, which stimulate the binding of Mediator proteins and, subsequently, RNA polymerase II (Pol II). The binding brings enhancer and promoter regions closer in space, and transcription starts. RNA Pol II proteins transcribe a gene into pre-mRNA and produce enhancer RNA (eRNA).

Around 70% of human promoters near TSSs contain a CpG island (CGI), which is a region with a high frequency of CpGs [23, 26, 27].

Based on promoter composition, nucleosome positioning, and what genes they regulate, promoters have four main types: type I, II, III, and IV [23]. Type I promoters are adult tissue-specific, and their transcription starts in a precise position. These promoters have a TATA box around 30 bp upstream from the TSS and have disordered along the sequence nucleosomes. Type II promoters control the expression of the housekeeping genes, and transcription initiation occurs at several locations. Type III promoters regulate developmental genes and also have a broad initiation pattern. These promoters depend on the input from distal enhancers. Type IV promoters are transcriptional machinery-specific and, like type I, have a sharp transcription initiation pattern. Type II, III, and IV promoters have a nucleosome-free region with precisely two nucleosomes positioned downstream and upstream. In addition, all three types of promoters have CGIs covering the TSSs [23].

2.1.2 Enhancers and silencers

Enhancers are distal regulatory regions that activate transcription by acting on a promoter over large genomic distances (Figure 2.5) [28]. The distance from the enhancer to the promoter it acts on can vary from several kilobases to megabases [13, 29]. The size of enhancers also varies from less than 100 bp long elements to 10–100 kb sequences, which are sometimes referred to as super-enhancers [28–30]. Enhancers contain one or more transcription factor binding sites. The various distribution of TFBSs within enhancers enables TF cooperativity [31], overviewed in the section about cooperative TF binding.

Enhancers often act spatially and temporally restrictedly, meaning that enhancers are cell-type specific and active only at particular time points [29]. Transcription factors approach enhancers and bind specific DNA patterns. Multiple TFs can bind one enhancer, depending on how many potential TFBSs that enhancer has [29, 32]. Transcription of the enhancer DNA produces non-coding RNA, termed enhancer RNA (eRNA) (Figure 2.5B) [28]. These RNAs promote the functions of enhancers by recruiting TFs, promoting enhancer-promoter looping, and influencing chromatin conformation or histone acetylation status [29, 32].

Enhancer-promoter looping occurs in isolated chromatin structures: loops and TADs (Figure 2.6). Cohesin is a ring-like structure that holds the chromatin loops together, letting enhancer and promoter communicate through complexes of proteins [28, 29, 33]. The formation of loops is known as loop extrusion. Another loop-extruding factor is protein CTCF. It binds its specific DNA motifs at the TAD boundaries, this way informing where to stop the loop extrusion process, closing the entire domain (Figure 2.6A) [33]. A single TAD consists of multiple loops held or not by cohesin [10, 13, 33] (Figure 2.6B-C). CTCF binding motif orientation is essential for the loop extrusion since two CTCF proteins bound convergently are required to close the loop [33]. The process insulates a particular region [29]. The influence of the enhancers is mostly, but not always, limited to the genes within the

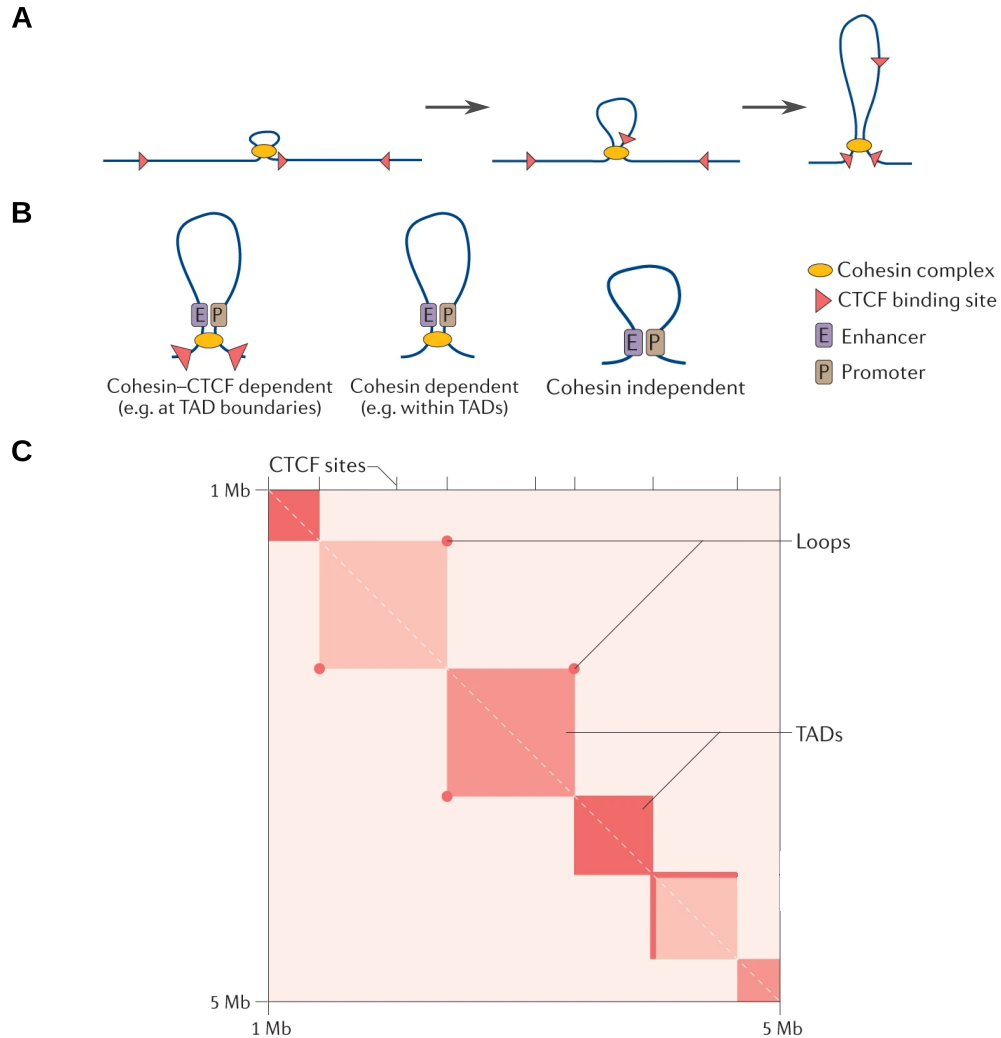


Figure 2.6: Chromatin is forming loops and topologically associated domains (TADs) using looping factors, such as cohesin and CTCF binding, to enable and spatially control promoter and enhancer communication. A. During loop extrusion, cohesin associates with DNA, and the loop is symmetrically extruded until cohesin encounters convergently oriented CTCF molecules. **B.** Loops can be cohesin and CTCF-dependent, marking the TAD boundaries (left). Within TADs, loops can be only cohesin-dependent (center) or entirely independent (right). **C.** A schematic heatmap represents the DNA-DNA contacts in a 4 Mb region in the mammalian chromosome. The more intense the color, the more frequent the contacts. TADs organize the contacts, and loops mark the TAD boundaries. Adapted from [33].

same loop [10, 13, 34]. One enhancer can regulate multiple genes, while multiple enhancers control one gene.

Opposite to the enhancers, the repressor proteins bind silencers to suppress the transcription [31]. Silencers can also be tissue-specific and operate within TADs through short- or long-range interactions [35, 36]. These regions are marked by specific chromatin marks, for example, Histone 4 Lysine 20 methylation (H4K20me) [35, 37]. Repressive TFs target silencers and, through looping, communicate with the gene's promoter [36].

2.2 Transcription factors

Transcription factors (TFs) are key transcription regulators [38, 39]. About ten percent of the human protein-coding genome codes for transcription factors [40]. TFs act as a bridge between the cell signaling pathways and gene expression. Through receptors, external cues trigger post-translational modification cascades that eventually result in the modulation of a set of TFs, which are then recruited at the gene cis-regulatory regions [39]. Depending on the stimuli, cell type, and state, TFs are modified. Short DNA sequences recognized by TFs are known as DNA motifs [41]. The function of TFs and their recognized DNA motifs are often studied together. Various experimental methods exist to identify TF-DNA interactions and various computational tools to model and represent those interactions. Most of the experimental techniques are limited to investigating individual TFs. However, in reality, TFs often cooperate, compete for the same binding site, or interact with the DNA indirectly [38]. These complex TF activity mechanisms are still challenging to investigate and interpret with currently available tools.

2.2.1 Protein structures and classification

Eukaryotic transcription factor proteins have at least two domains: DNA binding and activation or repression. Optionally, there are protein-protein interaction (PPI) or effector domains to regulate TF activity [42–44]. DNA-binding domain (DBD) recognize and bind specific DNA patterns (motifs). Many protein-DNA complex structures are available in resources, such as Protein Data Bank (PDB) [44–46]. For example, the basic helix-loop-helix (bHLH) domain proteins have two α -helices connected by a turn. These domains are known to dimerize, and each monomer recognizes one side of a symmetric DNA sequence (Figure 2.7A). bHLH proteins bind at the major groove of the DNA molecule [45]. Homeodomain-type DBDs are highly conserved. POU5F1, also known as OCT4, has two DBDs, which are binding cooperatively at the major groove (Figure 2.7B).

DBDs are extremely well-conserved and are used to classify TFs into families and classes. The similarity of amino acid sequences forming a DBD is the basis of the TF classification. TFs from the same class or family tend to recognize the same or highly similar DNA motifs (Figure 2.7C) [38]. It is an important feature that can help to define binding specificities of understudied TFs (a process called DNA binding motif inference) [44]. In addition, DBD can provide information

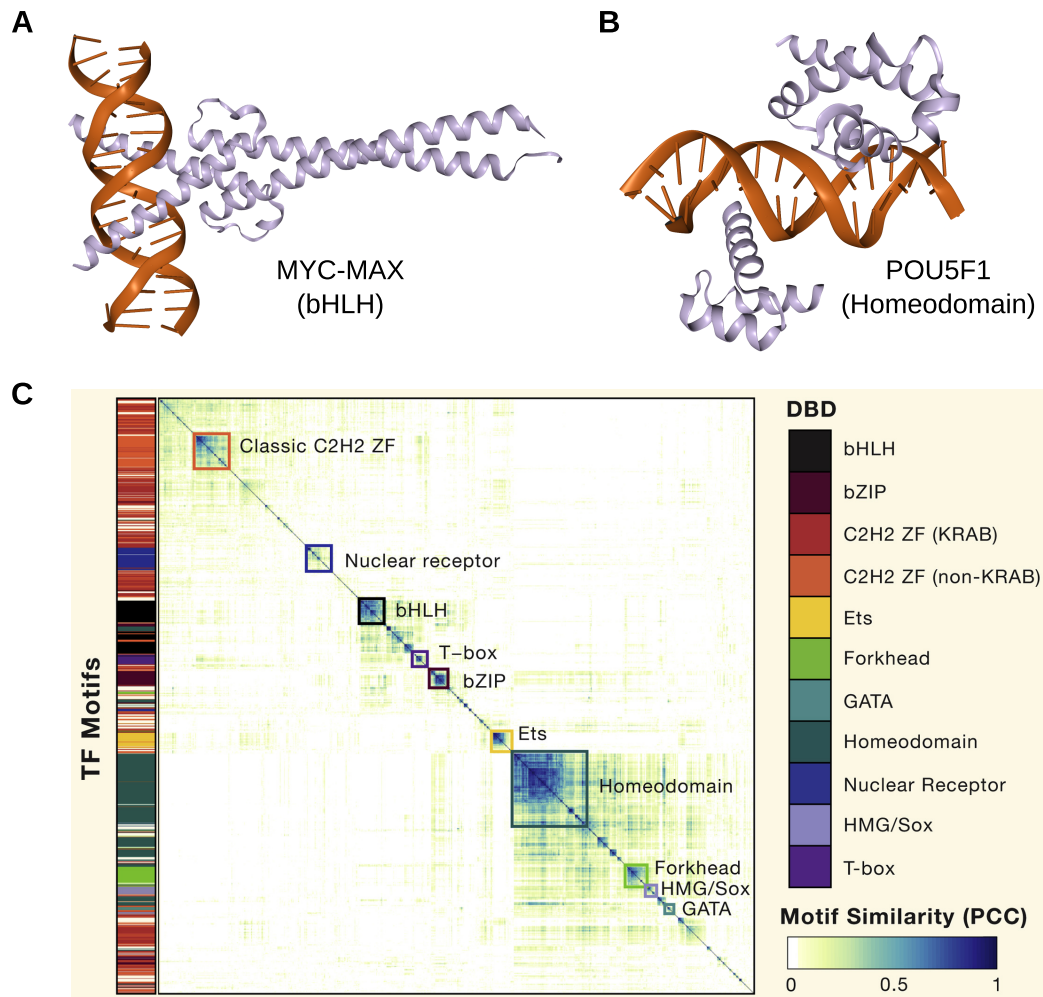


Figure 2.7: Transcription factors with the same DNA binding domains (DBDs) bind similar DNA sequence patterns. **A.** 3D model of *Homo sapiens* TF dimer MYC and MAX DBD binding to the DNA (PDB ID: 1NKP). MYC and MAX belong to a basic helix-loop-helix (bHLH) TF class. Proteins in purple, DNA in orange. **B.** Same as A, but *Homo sapiens* POU5F1 DBD interaction with DNA (PDB ID: 8G87). POU5F1 belongs to a homeodomain TF class. **C.** Often, TFs with similar DBDs are binding to highly similar DNA motifs. A heatmap represents human TF motif pairwise comparisons. Motifs group according to the protein DBDs. Adapted from [41].

about the protein function. For example, different homeodomain DNA binding domain-containing TFs often regulate developmental processes [47].

In 2009, a group of researchers defined and described 23 human TF families, where 80% of all known human TFs at the time had C₂H₂ zinc-finger, homeodomain, or HLH DBDs [47]. Later classifications expanded from humans to all mammalian TFs. One of the most extensive classifications of mammalian transcription factors, TFClass, includes superclass, class, family, subfamily, and genus categories [44, 48, 49]. For example, in the JASPAR database (storing TF-DNA binding computational models), TF families and class categories are the most used to annotate TF-DNA binding profiles. The most up-to-date version classifies more than 39,000 TFs from up to 41 mammalian species. The TFClass annotates TF DBDs and classifies TFs using phylogenetic analyses [48]. More recently, a similar classification of plant TFs was released. Many DBDs are present in plants but not mammals, which hindered the complete classification of the plant TFs. Plant-TFClass addressed that by adding eight new classes and 37 new families to classify most of the 56 plant TF types [50]. This recent development in TF classification highlights the need to consistently revise and update current classifications, also considering other taxa, such as nematodes or insects.

2.3 Transcription factor binding

Transcription factors bind cis-regulatory regions (promoters and enhancers) to regulate associated gene transcription. TFs typically recognize short (10-20 bp) DNA sequence motifs [51]. Depending on the sequence context and available co-factors, TFs can either activate or repress transcription of a gene [41]. Interactions with CRRs are highly cell-type-specific. The location where a TF binds DNA is called a transcription factor binding site (TFBS). A promoter or an enhancer can have multiple TFBSs of the same or different TF. Transcription factors generally have a binding preference for particular DNA patterns, also known as DNA motifs [52].

How do transcription factors find these small binding sites in a long DNA molecule surrounded by various proteins and molecules in the nucleus? The four primary forms of TF motion in the environment with DNA are (1) 3D Brownian diffusion in the nucleoplasm, (2) 1D sliding along the DNA, where TF binds DNA in a non-specific fashion, (3) already bound TF binds a nearby region (intersegmental transfer), (4) TF unbinds from a region, briefly diffuses, and rebinds to a nearby section of the same region (hopping) [53]. Each motion takes a different time for a TF to find a target region.

The way a transcription factor interacts with the DNA influences transcription regulatory mechanisms. TF binding varies from direct binding to the DNA (most well-studied) to indirect and cooperative binding [52, 54]. TF binding to the DNA depends on various features that co-exist together. For example, DNA modifications, such as DNA methylation, can dictate whether a TF will bind. DNA shape, presence of co-factors, or sequence variants can also determine TF binding [54]. It is essential to understand these mechanisms. The different binding

modes can increase the complexity of gene transcription regulation. It is necessary to ensure cell-type-specific transcription.

This section will overview direct, indirect, and cooperative TF binding, the impact of pioneer TFs, and DNA methylation. Various experimental techniques enabled the deciphering of TF binding mechanisms, and the section will briefly describe the most popular and widely used methodologies.

2.3.1 Direct and indirect TF binding

Direct and indirect TF binding is an often discussed topic. TF directly binds DNA when it interacts with the DNA in a sequence-specific manner. In such cases, the protein recognizes a specific DNA motif. However, the affinity of this interaction can vary. In other instances, TF can bind the DNA indirectly. It primarily describes the binding through the different proteins [55]. Some TFs can bind both directly and indirectly. For example, transcription factor Spi-1/PU.1 can bind to DNA directly and indirectly. Direct binding of Spi-1/PU.1 to the consensus motif *AGGAA* leads to transcriptional activation, while indirect binding through a partner GATA1 to the *GATA* motif may lead to transcriptional repression [52].

Experimental techniques that map TF-DNA interactions, such as ChIP-seq, potentially capture direct and indirect TF binding. Therefore, careful analysis of those regions and interpretation of the results are required. DNA motif analysis is the simplest way to determine whether the TF is binding directly (canonical motif present) or indirectly (another motif present) [55]. A section in the next chapter discusses computational approaches to distinguish direct from indirect TF-DNA interactions.

2.3.2 Cooperative transcription factor binding

Cooperativity (or co-binding) of TFs on regulatory regions, such as enhancers, is an essential transcription regulation mechanism [56]. Interaction with other sequence-specific or non-DNA binding co-factors can alter binding specificity [54]. Cooperative binding does not always mean that two TFs will bind the DNA directly; they can cooperate through protein-protein interactions or bind DNA indirectly [38].

We can analyze the cooperativity between TFs with DNA binding specificity through their binding DNA patterns. The rules to describe the cooperativity between the TFs are often referred to as motif grammar [56]. It is the presence of specific motifs, their order, the spacing between them, and the orientation of those motifs [56]. The two motifs can also overlap, thus creating a composite motif co-bound by a dimeric TF. Such cases are often challenging to investigate due to the limitations of experimental and computational tools. Some TFs require a rigorous motif grammar for their cooperativity. Others bind with various spacings, described as a relaxed grammar or soft syntax [57]. The orientation of the DNA motifs alone can be a strong determinant of transcription activity. For example, Spi-1/PU.1 modulates gene expression in erythroid cells and promotes oncogenesis. Research has also shown that it cooperates with KLF family TFs. When the

Klf-1 motif is in direct orientation in the promoter regions, Spi-1/PU.1 activates transcription, while the genes are repressed when Klf-1 binds the motif in reverse orientation [52].

Currently, three main models describe cooperative TF binding at the enhancer regions: enhanceosome, billboard, and TF collective model (Figure 2.8) [31, 56, 58]. The two main models are enhanceosome, which requires a rigid motif grammar, and the billboard model, which allows soft motif syntax and the presence of specific TFs is more important than the order, orientation, and spacing of the motifs (Figure 2.8A-B). Regulatory regions bound by TFs using an enhanceosome model are highly conserved in evolution, and TFs bind directly to strictly organized motifs. (Figure 2.8A). Precise binding site orientation, order, and positioning are required for an enhancer to function [58]. Regions following the billboard model are less conserved, and cooperativity is indirect, but TFs also bind directly to the DNA motifs (Figure 2.8B). The positioning, orientation, and distance of cognate motifs and their order vary without impacting the activity of the enhancer. The presence of the TF binding sites is the most important aspect [58]. The third model, TF collective, falls somewhere in between the first two (Figure 2.8C). In this case, the regions are also less conserved. TFs can bind directly and indirectly; motifs are not always present, and they have flexible organization. Finally, one enhancer can include a mixture of all models. Such regions have a variable sequence conservation, TFs can bind directly and indirectly, some motifs have a strict grammar, some have more flexible organization, or motifs are not present at all [56].

Understanding how different transcription factors work together to regulate transcription is essential. The cooperativity of TFs provides an incredible complexity of regulatory mechanisms, which allows spatial and temporal control of cell differentiation and growth. Specific experimental techniques, such as CAP-SELEX, have been developed to profile dimeric TF binding (see section experimental methods for TF-DNA interaction detection) but are still limited. Computational tools provide more opportunities to look into TF cooperativity. Simple computational models, such as position weight matrices, are not well suited to represent cooperative binding. However, deep learning techniques nowadays have a lot of potential in this area (read more about computational models in the next chapter).

2.3.3 Pioneering

A specific group of transcription factors can interact with the nucleosome-protected closed chromatin and trigger its opening for other TFs and proteins [56, 59]. These TFs are known as pioneer transcription factors or master regulators [56]. FOXA and GATA4 were the first factors detected to engage the enhancers in development, and their occupancy preceded occupancy of the other TFs that bind to those enhancers [60–62]. Well-characterized pioneer TFs include SOX2, KLF4, and OCT4, also known as Yamanaka factors, because they can induce pluripotency [62]. Pioneers bind the condensed chromatin before the transcription activation, therefore showing their essentiality for region transcriptional activation

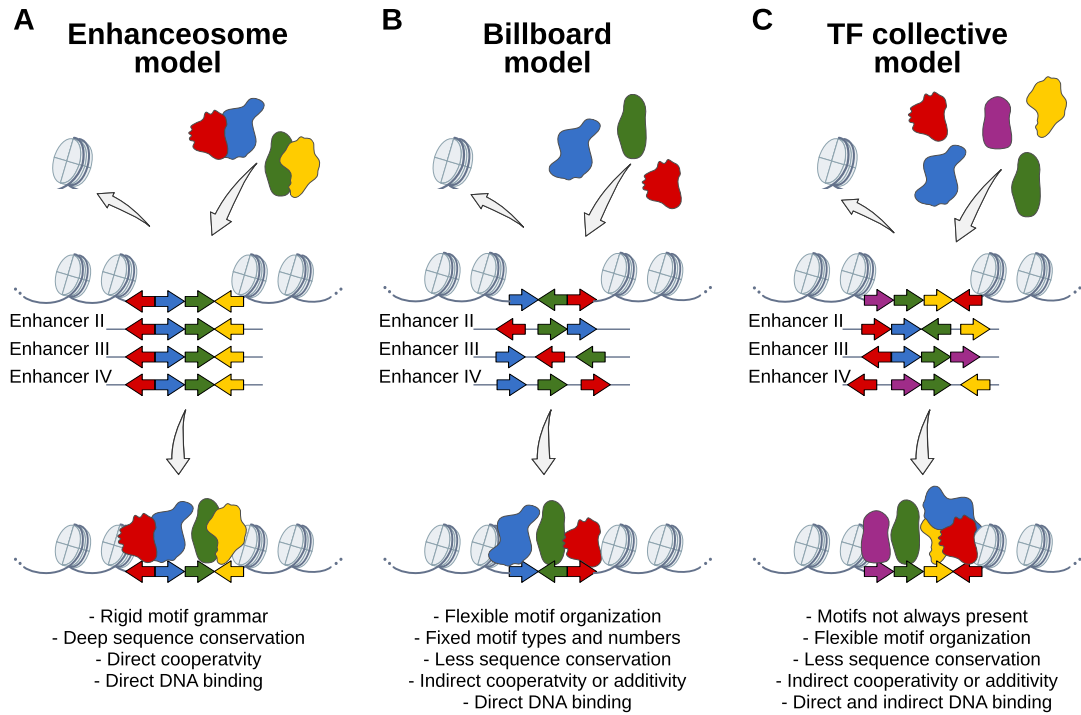


Figure 2.8: Enhancer models for TF cooperative binding. **A.** Cis-regulatory regions that follow the enhanceosome model have rigid motif grammar and deep sequence conservation and are bound directly by directly cooperating TFs. **B.** Regions that follow the billboard model have a much more flexible motif organization but fixed motif types and numbers. These regions are less conserved but also directly bound. In contrast to the enhanceosome model, TFs are cooperating indirectly or additively. **C.** TF collective model has features from both enhanceosome and billboard models. Sequences are less conserved, and motif organization is flexible. DNA motifs are not always present, and TFs can directly and indirectly interact with the DNA. Adapted from [56].

[62]. In some cases (p53) (SOX2, KLF4, and OCT4), it is enough for a pioneer TF to recognize a partial DNA motif to bind closed chromatin; in other scenarios, pioneers prefer binding near the nucleosome edge, and their binding affinity is more substantial than to the sites located in the naked DNA [62]. After binding, pioneer TFs open closed chromatin locally first. Then ATP-dependent chromatin remodelers, such as the SWI/SNF remodeling complex, are recruited to promote nucleosome eviction in the region [59]. Finally, acetyltransferases modify histone tails and activate the cis-regulatory element [52, 59]. In cancers, some pioneer TFs promote DNA region demethylation [63]. Aberrant methylation modulates chromatin structure and transcription of cancer-related genes.

2.3.4 Impact of methylation to the DNA binding

DNA modifications impact how transcription factors bind [64]. The most extensively studied epigenetic DNA modification is DNA methylation. “Writers”, DNA methyltransferases (DNMTs), modify the cytosines in the DNA molecule by adding a methyl group in the context of CpG dinucleotides, generating 5-methylcytosine (5mC) (Figure 2.9A). In contrast, ten eleven translocation (TET) enzymes conduct the removal of methyl groups [64, 65]. Generally, hypermethylation of CpGs in the promoter regions is associated with transcription repression, while gene body hypermethylation indicates active genes [65, 66]. Therefore, DNA methylation is essential for TFs that bind methylated motifs, and some prefer to interact with methylated sites [65, 67]. For example, the HOXB13 protein has three key residues that contribute to the preferential mCpG binding (Figure 2.9B-C) [67]. HOXB13 can also bind to the unmethylated motif, which then does not contain a CpG dinucleotide (Figure 2.9C). Some TFs (for example, MAX) are not influenced by methylation even though their motifs contain a CpG [67]. For other TFs, DNA hypermethylation suppresses their binding. Changes in the methylation patterns can lead to aberrant TF binding, affecting transcription regulation and possibly causing diseases.

Most common TF-DNA interactions computational models do not consider methylated CpGs and position interdependency. The dependency between the positions in the TF-recognized DNA sequence is of particular relevance when analyzing CpG dinucleotides. A CpG in the binding motif does not inform whether it needs to be methylated for the TF to bind. There are specific attempts to model TF binding to the methylated DNA. For example, the MEDEMO toolkit analyzes TF binding motifs by including information about DNA methylation [68]. The simplest way to include methylation information is to extend the alphabet where 5mC is replaced with M and G on the opposite strand with an H [68, 69]. Methylation-aware TF binding models can use such methylation-aware genomes for training. These can, therefore, inform about methylation sensitivity (Figure 2.9D) and allow the identification of differential TF binding regions in ChIP-seq data [68]. The modeling can also include other DNA modifications (5-hydroxymethylcytosine, 5-formylmethylcytosine, or 5-carboxymethylcytosine) by coding them as specific letters [69].

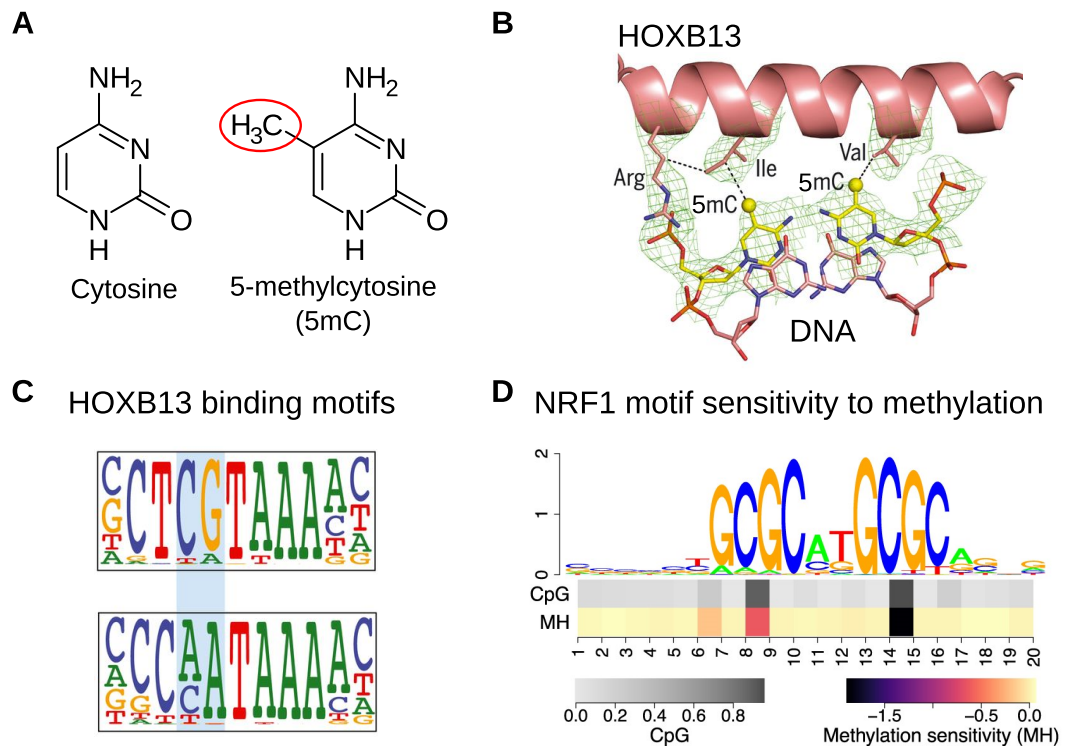


Figure 2.9: TF binding is affected by DNA methylation. **A.** DNA methyltransferases add methyl- group to the cytosines in the CpG context and generate 5-methylcytosine (5mC). **B.** Some TFs, such as HOXB13, specifically recognize DNA sequences with methylated cytosines. Three key protein residues, Arg, Ile, and Val, direct the specificity towards 5mC. **C.** HOXB13 prefers binding to the motif containing a methylated CpG (primary motif; above) but also binds a motif without a CpG with lower affinity (secondary motif; below). **D.** Some specific methodologies investigate TF binding sensitivity to methylated sites. For example, NRF1 is sensitive to methylation and binds to a motif with CpG content. Based on the sites bound by NRF1, NRF1 does not prefer methylated CpGs. The first CpG is less affected by methylation than the second. Adapted from [67, 68].

2.3.5 Experimental methods for TF-DNA interaction detection

Over the years, scientists developed multiple experimental methods to detect the binding preferences of the transcription factors. Broadly, these methods are applied *in vitro* or *in vivo*. Most of the techniques, though, are limited to investigating the interactions of a single TF at the time. This section will describe the most popular and relevant methods. In addition, it will explain a few techniques that probe open chromatin. Information on the open chromatin can also define the transcription factor binding landscape.

In vitro

In vitro methods mainly aim to identify TF binding sequence preferences. For example, systematic evolution of ligands by exponential enrichment (SELEX) or PBM. Other methods, such as EMSA, can inform about the binding affinity [70]. The most significant limitation of *in vitro* methods is that they cannot give information about the biological context of the TF-DNA interaction by design.

SELEX is a qualitative method developed decades ago, but high-throughput (HT-SELEX) and other versions are extensively used today [71, 72]. The basic principle of SELEX is the incubation of a purified TF with a pool of random DNA oligonucleotides (Figure 2.10A). Random sequences bound to the protein are collected, amplified, and incubated with the TF again. After multiple cycles of incubation, separation, elution, and amplification, the sequences with the highest affinity are selected and sent to the sequencing procedure. The resulting sequencing data is analyzed to determine the TF binding sequence preference [70, 72]. Differently from a regular SELEX, one can study multiple transcription factors simultaneously in HT-SELEX. However, since the oligonucleotides are random, the sequences bound with the highest affinity in SELEX (or HT-SELEX) are not guaranteed to reflect the functional binding preferences in a living cell.

One variation of the SELEX protocol is particularly relevant because it allows for studying dimer TF sequence preferences. In the consecutive affinity-purification systematic evolution of ligands by exponential enrichment (CAP-SELEX) protocol, oligonucleotides are incubated with two TFs, flagged with different flags. During two consecutive affinity purification steps, only those sequences bound by both TFs proceed to amplification [73]. This technique shows how specifically two TFs can bind the same sequence. For example, we can determine preferred spacings or overlaps between two individual motifs and motif orientation [73].

A high-throughput technique falling in between *in vitro* and *in vivo* is called DNA affinity purification sequencing (DAP-seq). DAP-seq uses native DNA to construct the libraries of sequences, preserving cell-specific modifications. Native DNA libraries provide increased specificity compared to methods such as SELEX. The library of sequences is then incubated with *in vitro*-expressed and tagged TF. TF-bound sequences are purified and sequenced. Finally, sequences are mapped to the origin genome to identify locations preferentially bound by a TF [74].

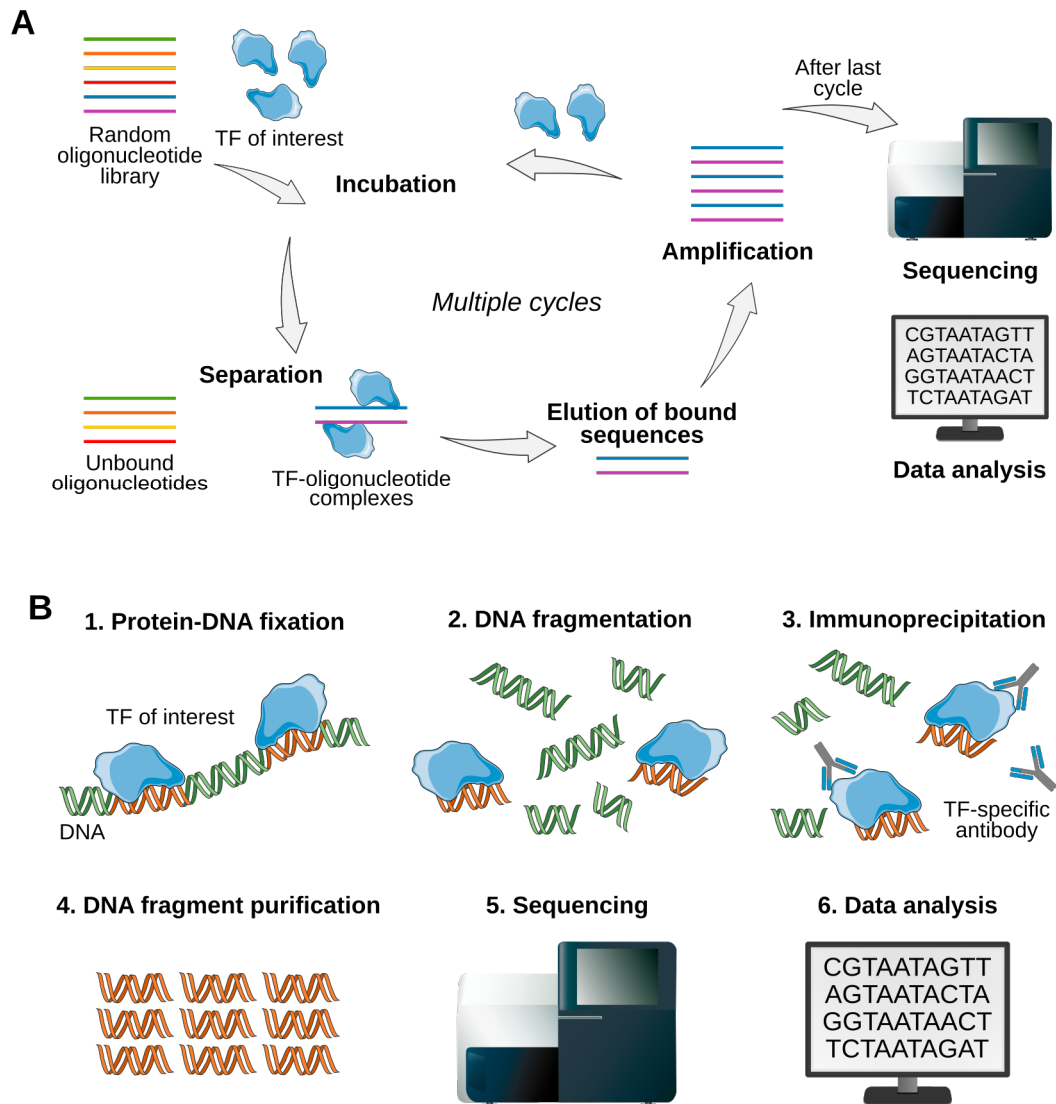


Figure 2.10: Examples of experimental techniques to investigate TF-DNA interactions. **A.** SELEX is an *in vitro* method that allows the determination of the TF binding sequence preferences. In the procedure, a pool of random oligonucleotides is incubated with the TF of interest. Unbound oligonucleotides are separated from TF-DNA sequence complexes and washed away. Bound sequences are purified, amplified, and incubated again. The incubation cycle is repeated several times, and after the last cycle, sequences bound with the highest affinity are sequences, and the resulting data is analyzed. **B.** ChIP-seq is an *in vivo* technique for identifying genomic locations bound by a TF of interest. DNA and proteins are cross-linked (1), and DNA is fragmented (2). The protein of interest is immunoprecipitated using a specific antibody (3). These DNA-protein complexes are selected, and DNA is purified (4). Eluted DNA is sequenced (5), and the resulting data is analyzed (6).

In vivo

Arguably, the most popular technique to map TF-DNA interactions in living cells is chromatin immunoprecipitation followed by sequencing (ChIP-seq) [75]. The ChIP-seq process starts with the cross-linking of the DNA and proteins in the living cells using formaldehyde (Figure 2.10B). Then, DNA is randomly fragmented with sonication, leaving DNA segments bound and unbound by proteins. In a crucial step, a specific antibody binds the TF of interest. The washing removes everything that is not TF-DNA complexes bound by antibodies. Further steps reverse protein-DNA cross-links and purify DNA fragments previously protected by the TF. These DNA sequences proceed to sequencing. After sequencing, sequencing reads are mapped to a genome. Further analysis identifies those genomic regions enriched by TF binding [75, 76].

The most significant advantage of *in vivo* techniques, such as ChIP-seq, is that the TF binding locations are determined in a specific tissue or cell line, thus reflecting the TF binding preferences in physiological conditions [76]. However, this comes with a few limitations. TF binding and regulatory mechanisms are complex, so *in vivo* binding is influenced by several factors that affect the outcome of the experiment. Differences in TF-DNA affinity, protein-protein interactions, indirect binding, and sequence biases can all affect how results are interpreted [72, 77, 78]. Other limitations of the ChIP-seq technology include studying only one TF at a time, and the immunoprecipitation requires a high-quality antibody, which can be difficult to acquire if the protein of interest is relatively unknown or poorly annotated [76].

In ChIP-seq, the DNA fragment sizes are variable and are longer (hundreds of base pairs) than the actual TF-protected region because DNA shearing is random [77, 79]. Therefore, the TF binding site inference is not in base pair resolution. Computational analysis and motif search cannot identify all binding locations in ChIP-seq derived regions [77, 80]. Researchers developed the ChIP-exo technique to overcome this limitation. The protocol has an additional step of immunoprecipitate treatment by 5' to 3' lambda exonuclease, which digests the unprotected ends of the DNA fragment bound by a TF. It allows us to get precise, nearly single-base resolution genomic locations bound by a TF [77, 80]. The length of such regions is 36-100 bp [79]. A simplified ChIP-exo protocol, ChIP-nexus, was developed in 2015 [78, 79].

The latest methods for TF binding region identification *in vivo* include CUT&Tag and CUT&RUN [81]. Both techniques are similar and are based on specific antibodies recognizing the TF of interest. In CUT&Tag, the antibody tethers the A-Tn5 transposase fusion protein. After its activation, transposase adds adapters for paired-end DNA sequencing, and high-resolution fragment libraries are generated [81, 82]. In CUT&RUN protocol, micrococcal nuclease (MNase) is tethered by the antibody and, when activated, cuts the DNA at the borders of TF binding [83]. These novel methods allow for small numbers of cells as input and profiling of TFs in single-cells (CUT&Tag is better suited for this application) and promise a more accurate determination of the TF-bound genomic locations by providing high resolution and exceptionally low background [81–83].

Even with new technologies emerging every year, ChIP-seq remains an essential

method for TF binding profiling. Many ChIP-seq data are publicly available, but researchers have not fully explored it. It motivates researchers to work on novel computational methods that allow them to study this data further.

Profiling chromatin accessibility

A separate category of experimental techniques aims to investigate open chromatin. Instead of profiling a single transcription factor, techniques such as assay for transposase-accessible chromatin using sequencing (ATAC-seq) or DNase I hypersensitivity (DHS) provide information on regions accessible to proteins, such as TFs.

In the DHS assay, a DNase I enzyme selectively digests nucleosome-depleted bulk DNA [84]. After digestion, DNA fragments are sequenced, and then reads are mapped to the genome. The computational analysis identifies DHS footprints that indicate the nucleosome-free regions, potentially bound by various TFs [84, 85].

ATAC-seq uses a Tn5 transposase to cut and insert sequencing adapters into regions unprotected by nucleosomes or other proteins [86]. The resulting DNA fragments are then sequenced. Sequencing reads are mapped to the genome. From reads, open regions (peaks) are called and analyzed computationally. Computational tools predict the binding of different TFs (more on TF binding site predictions in the next chapter) [86]. ATAC-seq can be in bulk or single-cell form. Investigation of open chromatin in single cells is used to define the cell-type specific accessibility, where TFs are accessing different regions [86].

A more recent technique, single-molecule footprinting (SMF), allows probing chromatin accessibility and co-occupancy of transcription factors in single-molecule resolution [87, 88]. The method utilizes DNA methylation to identify genomic locations not protected by transcription factors and nucleosomes. DNA is subjected to methylation in CpG or GpC contexts and then sequenced using bisulfite sequencing. It provides information on methylation status for each molecule/read [87, 88]. The methylation status can be summarized for each region of interest, for example, a region containing the potential binding sites for two TFs (anchor and co-binding TFs). The more methylated, on average, the genomic region is, the more accessible it is (Figure 2.11A). [87, 88]. Further computational analysis infers five occupancy states for each molecule and region: “Co-binding + Anchor”, “Co-binding”, “Anchor”, “Accessible”, and “Nucleosome”. Bins, overlapping and neighboring regions of interest, summarize the methylation of individual CpGs. This binarized data represents different occupancy states (Figure 2.11B).

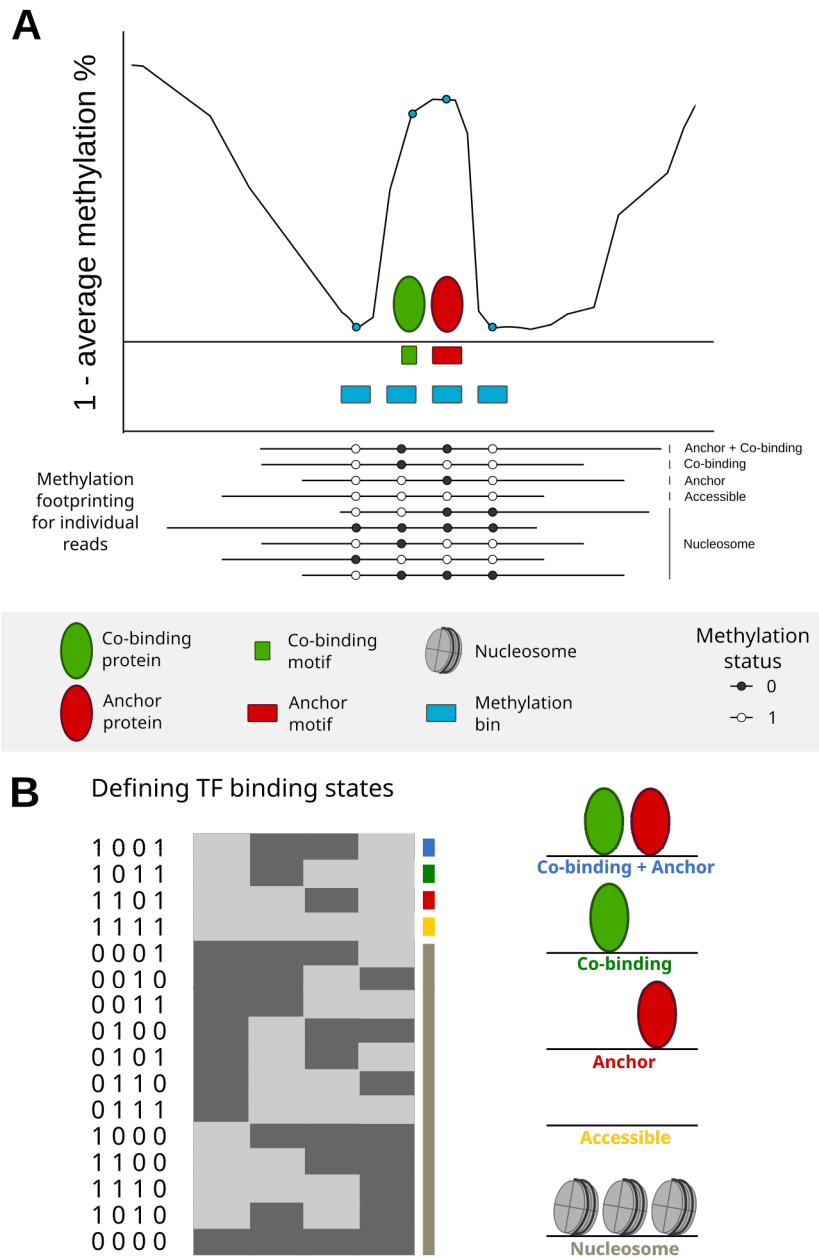


Figure 2.11: Single-molecule footprinting (SMF) technique enables investigation of cooperative events on single-molecules. **A.** SMF results in methylation information for single molecules. Methylation can then be analyzed and summarized for individual regions of interest (for example, the potential binding site for anchor and co-binding TFs, red and green blocks, respectively). The average methylation signal visualized as a line plot gives a general overview of the accessibility, where the lack of methylation represents open chromatin. From SMF data, the occupancy states can be inferred by summarizing methylation in four bins that overlap with regions of interest and neighbor them (blue blocks). **B.** From binarized data, five states for each molecule can be inferred: “Co-binding + Anchor”, “Co-binding”, “Anchor”, “Accessible”, and “Nucleosome”. Adapted from [89].

